Med-HEAL: Analyzing and Mitigating Hallucinations in Medical LLMs with Hallucination-Aware In-Context Learning

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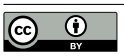
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Abstract

Hallucinations in medical large language models (LLMs) pose serious risks for clinical decision support, particularly when models must reason over complex electronic health records (EHRs). However, existing benchmarks often lack a realistic clinical context and provide limited insight into how hallucinations can be mitigated in practice. We introduce Med-HEAL, a framework for systematically identifying, analyzing, and mitigating hallucinations in medical LLMs using clinically grounded data. Building on the EHRNoteQA benchmark derived from MIMIC-IV discharge summaries, we construct a hallucination dataset by evaluating BioMistral-7B on open-ended clinical question answering tasks. Model outputs are labeled through a dual evaluation pipeline that combines LLM-as-a-Judge assessment (GPT-4o) with human auditing by medical student reviewers, producing correctness judgments and annotations of reasoning errors via a custom web-based evaluation system.

We then leverage this dataset to investigate mitigation strategies: a self-critique pipeline, in which the test model reviews its own answers to detect potential errors and regenerates responses for flagged cases, and retrieval-augmented in-context learning (RA-ICL), which exposes the model to hallucinated and corrected examples. Experiments across five open-source LLMs—BioMistral, Llama-3.1, DeepSeek, Qwen2.5, and Qwen3, show that the self-critique strategy improves accuracy for three of five models ($p < 0.05$) without requiring parameter updates.

Med-HEAL provides both a reusable hallucination dataset and a practical framework for studying and mitigating hallucinations in medical LLMs, supporting safer deployment of AI systems in clinical environments. Our code and data are publicly available at https://github.com/yimingliao-blad/med-heal.git.



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CCS Concepts

• **Computing methodologies** → **Few-shot learning**; Natural language processing; • **Applied computing** → *Bioinformatics*.

Keywords

Large language models, Medical AI, Hallucination detection, In-context learning, Electronic health records, Clinical decision support

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1 Introduction

The integration of large language models (LLMs) into healthcare has the potential to transform how clinicians interact with medical data. Modern healthcare systems generate vast amounts of information through electronic health records (EHRs), which contain rich but largely unstructured clinical narratives [21]. LLMs provide powerful capabilities for analyzing such data, enabling automated information extraction from clinical notes, summarization of patient histories, and assistance with clinical reasoning tasks [3]. By reducing the burden of navigating complex documentation, these systems may help alleviate the growing workload and information overload faced by healthcare professionals.

Recent research has adapted general-purpose LLMs to the medical domain using domain-specific corpora [21], instruction tuning [3, 16, 22], and retrieval-augmented generation techniques [15, 26]. The long-term vision is to deploy LLMs as on-demand clinical assistants that can analyze a patient's medical history, answer clinicians' questions, and support diagnostic decision-making.

Despite these promising capabilities, the deployment of LLMs in healthcare remains constrained by the problem of *hallucinations*—the generation of incorrect, fabricated, or unsupported medical information. While hallucinations in general-purpose applications may cause minor inconvenience, errors in clinical contexts can lead to serious consequences. Detecting such errors is particularly challenging because evaluating clinical reasoning often requires domain expertise. Recent studies have highlighted the prevalence of hallucinations in medical LLMs and proposed benchmark datasets to measure these failures [2, 11, 18].

However, several important challenges remain.

- **Non-clinically-realistic hallucinations in existing medical hallucination benchmarks.** Many current benchmarks separate clinical questions from the complex contexts in which healthcare decisions are made. Datasets such as MedHalu [2] and MedHallu [18] draw questions from structured knowledge bases including PubMedQA [10], LiveQA [4], USMLE [9], and MedicationQA [1]. While these sources provide high-quality medical questions, they primarily test factual recall rather than the reasoning required to interpret patient records. Moreover, many benchmarks generate synthetic hallucination examples. For example, MedHallu [18] iteratively refines incorrect answers using GPT-4 to produce adversarial distractors. Such artificially constructed errors may differ substantially from the spontaneous reasoning failures that occur when LLMs analyze real clinical narratives.
- **Limitations of fine-tuning-based mitigation methods.** Fine-tuning has been widely used to reduce hallucinations in medical LLMs. For instance, MedCaseReasoning collects diagnostic reasoning cases from clinical reports and performs supervised fine-tuning to reduce logical errors in model outputs [24]. However, fine-tuning presents several practical challenges. First, medical knowledge evolves rapidly as treatment guidelines, drug approvals, and clinical protocols are updated, which can quickly render fine-tuned model parameters outdated. Second, repeated fine-tuning may introduce catastrophic forgetting [8, 12] or degrade general reasoning ability, potentially creating new forms of hallucination [7].

Scope of This Work. To better understand and mitigate hallucinations in clinical reasoning tasks, we introduce **Med-HEAL**, a framework for collecting realistic hallucination examples and using them to improve model reliability at inference time.

First, we construct a clinically grounded hallucination dataset by leveraging the existing dataset EHRNoteQA [13]. We utilize the error outputs of BioMistral-7B, a fine-tuned medical-domain LLM, and evaluate them with a dual-judge pipeline that combines LLM-as-a-Judge assessment with human auditing, enabling scalable labeling while maintaining clinical fidelity.

Second, we propose an inference-time mitigation framework centered on self-critique. In this pipeline, the model first generates an answer, then critiques its own response to detect potential reasoning errors, retrieves relevant demonstrations from the Med-HEAL dataset, regenerates the answer when errors are identified, and finally verdicts whether to accept the corrected answer only when it is better than the original answer. We also investigate plain retrieval-augmented in-context learning (RA-ICL) using relevant hallucination examples from the Med-HEAL dataset. Our experiments show that while RA-ICL alone provides limited benefits, the self-critique mechanism consistently improves performance over zero-shot answers across multiple models with statistical significance.

In summary, this work makes the following contributions:

- We introduce **Med-HEAL**, a framework for collecting clinically grounded hallucination examples from LLM interactions with real EHR discharge summaries.
- We propose an inference-time mitigation pipeline based on self-critique, enabling models to detect and correct reasoning errors without parameter updates.
- Through experiments across five open-source LLMs, we demonstrate that self-critique consistently improves clinical question answering accuracy on EHRNoteQA, outperforming retrieval-based ICL and chain-of-thoughts prompting.

2 Methods

We will start with an overview of the components of the Med-HEAL framework, then describe the key contributions for each component.

2.1 Overview

We introduce **Med-HEAL**, a framework for systematically identifying and mitigating hallucinations in medical large language models (LLMs). The framework contains two core components:

(1) Hallucination Data Collection Pipeline, which extracts clinically realistic hallucination cases from LLM outputs using a dual-judge evaluation system combining LLM-based evaluation and human auditing.

(2) Self-critique and retrieval-augmented In-Context Learning, which incorporates hallucinated reasoning pinpoint and corrected format examples into retrieval-augmented prompts to guide LLM responses.

The overall pipeline is illustrated in Fig. 1. First, a medical LLM generates answers to EHR-based clinical questions. These outputs are evaluated by both an automated LLM judge and human reviewers to identify hallucinations and collect explanatory annotations. The resulting dataset is used to construct contrastive demonstrations for in-context learning on downstream models.

2.2 Hallucination Data Collection Pipeline

Clinical QA Generation. The initial hallucination data used in this study were generated by running BioMistral-7B [14] on the 962 open-ended questions from EHRNoteQA [13]; additional EHRNoteQA details and topic coverage are provided in supplementary materials. We select BioMistral because it is a representative, customized medical LLM that is prone to generating hallucinatory results, making it useful for hallucination data collection. BioMistral is based on the Mistral-7B-Instruct model, fine-tuned with a vast corpus of medical research papers from PubMed Central for complex medical question-answering and clinical reasoning tasks.

We use the open-ended setting because it exposes free-form reasoning errors that are central to hallucination analysis. Due to

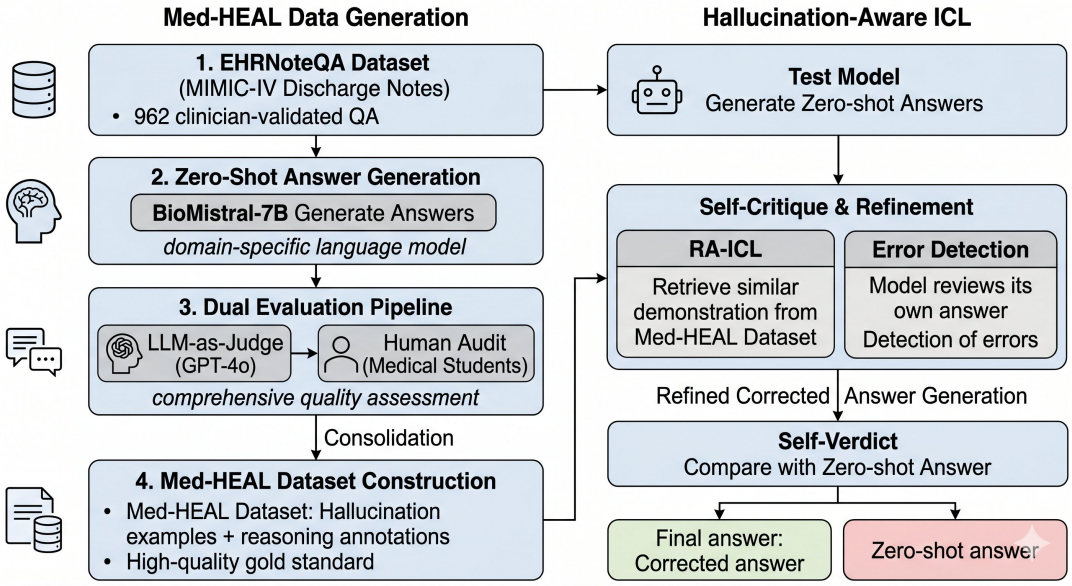


Figure 1: Overall Pipeline of Med-HEAL. Phase 1 (Left): LLM-generated labels are audited against human ground truth provided by medical student reviewers; these annotations are then consolidated to construct the hallucination dataset. Phase 2 (Right): The Self-Critique pipeline performs error detection and utilizes retrieval-augmented in-context learning (RA-ICL) with similar demonstrations to generate refined answers, which are then finalized through a self-verdict step.

the workload of reviewing long discharge-summary contexts, the current human validation set is limited to 100 samples.

Dual-Judge Hallucination Identification. Med-HEAL uses a dual evaluation system that combines GPT-4o judging with human auditing. GPT-4o first compares each BioMistral answer against the EHRNoteQA ground truth to produce a match/mismatch label, and medical student reviewers audit a 100-sample subset to validate the judge.

To improve the efficiency of human evaluation, we developed a web application that presents the same discharge note(s), ground-truth answer, and BioMistral answer to both LLM and human judges. Human disagreements are resolved by majority vote.

We further analyze how these hallucination examples are distributed across categories derived from prior work [11]. The detailed taxonomy table and examples are provided in supplementary materials. Because full human review of error rationales is resource-intensive, we use GPT-4o with category exemplars in the prompt to classify hallucination examples. Table 1 shows the resulting distribution, compared with reported error distributions in prior studies [11, 13].

Med-HEAL Dataset Construction. The Med-HEAL dataset is constructed by gathering BioMistral model outputs on the EHRNoteQA benchmark. Specifically, each entry contains $r = (D, q, a, a^*, y, e)$, which include the EHRNoteQA context (discharge summary D), the clinical question q, the BioMistral-generated answer a, the ground truth answer a^*, the LLM judge's output $y \in \{0, 1\}$, where $y = 1$ indicates a hallucination case, and the main error e extracted from a when $y = 1$. Unlike synthetic hallucination benchmarks, Med-HEAL captures naturally occurring hallucinations from realistic EHR reasoning tasks, enabling the analysis of real-world failure modes.

To facilitate efficient retrieval of correct and hallucination examples, each component of the entry, i.e., D, q, a, a^*, and e, is vectorized for similarity-based retrieval. After testing a set of similarity-based retrieval measures, such as BM25 [19], KATE [23], and GTR-T5-Base encoding [17], we adopt GTR-T5-base encoding, as it yielded the best retrieval performance for semantic similarity when applied to RA-ICL.

2.3 Hallucination-Aware In-Context Learning

To mitigate clinical reasoning errors, we introduce **hallucination-aware in-context learning (ICL)**. This method utilizes documented hallucination instances from the Med-HEAL Dataset to guide the correction of detected errors, thereby reducing overall hallucination rates. The framework includes the following core components:

Self-Critique. We employ a self-critique and retrieval framework to select contrastive few-shot cases that guide the downstream model in correcting its own reasoning, thereby producing a hallucination-free response.

The self-critique framework consists of two steps: error detection and correction. Given a test case (D_t, q_t), the target model first generates a zero-shot answer a_t. In the first detection step, given D_t, q_t, and a_t, the target model will try to identify the most significant error, denoted by e_t. If the detection finds no error, we accept the zero-shot answer a_t without any changes. Otherwise, we will proceed to the correction step.

The error correction step uses the detected error e_t to retrieve a contrastive few-shot case from the Med-HEAL lookup set. Given the current case (D_t, q_t), we query hallucination records, i.e., $r_i = (D_i, q_i, a_i, a_i^*, y_i, e_i)$ with $y_i = 1$, and score each candidate by discharge-summary similarity:

$$\text{sim}(D_t, D_i) = \cos(\phi(D_t), \phi(D_i)),$$

Here, $\phi(\cdot)$ denotes GTR-T5 encoding, which converts the given discharge-note text into an embedding representation; the record with the highest similarity score is retrieved as the most relevant case.

Self-Verdict. Finally, a self-verdict mechanism using a prompt similar to the detection step in self-critique to detect potential errors in the refined answer. If the self-verdict indicates that the correction leads to more errors than the zero-shot answer a_t, the system rejects the revision and retains a_t. Otherwise, the revised response is accepted as the output of the whole **hallucination-aware in-context learning (ICL)** pipeline.

2.4 Experiment Setup

Hardware and Software Environment. Our experiments were conducted on a local workstation equipped with an Intel Core i9-13900K processor (24 cores, 32 threads), 251 GB RAM, and a single NVIDIA GeForce RTX 4090 GPU (24 GB VRAM), running Ubuntu 24.04 LTS. Local models were served via the vLLM framework for efficient inference. All GPT-4o API calls for the dual-judge evaluation were made through the official OpenAI API. Runtime comparison details are provided in supplementary materials.

Baselines. We compare hallucination-aware prompting with several baseline prompting strategies, including zero-shot prompting, positive retrieval (correct examples only), negative retrieval (hallucination examples with corrections), naive contrastive (one correct and one hallucination example) retrieval, and Chain-of-Thought prompting. Detailed prompt condition descriptions are provided in supplementary materials.

Unlike parameter fine-tuning approaches, the proposed method transfers hallucination knowledge entirely through prompt demonstrations. This allows the framework to be applied to multiple open-source LLMs without retraining.

ICL data split. For the retrieval-augmented ICL (RA-ICL) experiments, we use 5-fold set. Each round, one-fold is used as the retrieval example set and tests on the other four folds.

Tested local medical LLMs serve different roles in the pipeline. BioMistral-7B is used as the source model to collect hallucination cases. Qwen2.5-7B-Instruct is used as the pilot model for designing the transfer ICL architecture, and the finalized setup is then evaluated across the full model set.

In-context learning: We evaluate Qwen2.5-7B [25], Qwen3-8B, LLaMA-3-8B [6], and DeepSeek-R1-Distill-LLaMA-8B [5], served via vLLM (temperature=0.1, maxtokens=8,096), under 5-fold cross-validation on 962 questions from EHRNoteQA. Answer correctness is assessed by GPT-4o as binary (yes/no), comparing against the ground-truth option text.

All prompts follow the ChatML format with a system message and a user message; the shared user prompt contains `Discharge Summary` and `Question` fields, with additional prompt details provided in supplementary materials.

The system message varies by condition; the base instruction tells the model to answer questions about discharge summaries as a medical expert.

For ICL conditions, demonstration examples and instructions are appended to the system message. Detailed condition descriptions are provided in supplementary materials. We systematically varied all conditions, including retrieval based on note similarity, question similarity, question-type matching, and random selections. For the composed cases, we included scenarios with only correct examples, only incorrect examples, contrasted examples (correct and incorrect), or random selection.

3 Results

3.1 Generated Med-HEAL Dataset

BioMistral-7B achieves an overall accuracy of 63.8% in a zero-shot setting on the 962 open-ended EHRNoteQA questions, with 348 incorrect answers that constitute the Med-HEAL hallucination set.

Answer correctness is determined through a two-stage evaluation pipeline.

(1) GPT-4o acts as a judge that determines whether the generated answer conveys the same key clinical facts/conclusions as the clinician-verified ground truth given by EHRNoteQA, producing a binary correct/incorrect label.
(2) GPT-4o analyzes incorrect answers to identify hallucination types, assigning scores on a 1–5 confidence scale for three categories: incomplete reasoning, factual errors, and fabricated sources, as discussed previously [11]. Scores of ≥ 4 indicate that a hallucination type is likely present.

The final dataset contains both the LLM-as-a-Judge and human reviewers' output and reasoning annotations; representative qualitative examples are provided in supplementary materials. Table 1 summarizes the results. Accuracy declines substantially as the number of discharge notes increases: 74.1% for one note, 62.3% for two notes, and 38.9% for three notes. This trend suggests that the model struggles to synthesize information from longer and more complex clinical narratives.

We apply GPT-4o to categorize the incorrect answers into three hallucination types. Each answer might be labeled with multiple error types. 98.0% incorrect answers are labeled with factual errors, followed by 82.2% with incomplete reasoning, and 5.2% with fabricated sources. These findings indicate that hallucinations in this task primarily arise from failures to integrate multiple pieces of evidence from patient records, rather than from inventing entirely unsupported facts.

3.2 LLM Judgment Calibration

To evaluate the reliability of our automated assessment, we benchmarked LLM-based judgments against independent human reviews from three medical students (A, B, and C). Reviewers A and B had completed their core clinical rotations, while Reviewer C was a first-year preclinical student.

Initial inter-rater reliability among the three reviewers showed fair-to-moderate agreement ($\kappa = 0.287$). This was primarily due

Table 1: BioMistral-7B zero-shot performance judged by GPT-4o on EHRNoteQA (962 questions). Hallucination types are among the 348 incorrect answers. (Confidence Level ≥ 4).

Metric	Count	Rate
Total questions	962	—
Correct answers	614	63.8%
Incorrect answers	348	36.2%
Hallucination type (multi-labels, among 348 errors, classified by GPT-4o)		
Factual errors	341 / 348	98.0%
Incomplete reasoning	286 / 348	82.2%
Fabricated sources	18 / 348	5.2%

to Reviewer C's systematically stricter labeling threshold, which lowered overall consistency. To mitigate this variance, we resolved conflicts via majority vote across a randomly sampled 100-item subset to establish a gold standard for comparison.

Table 2 reports accuracy between each judge and the gold standard. Reviewers A, B, and C have 89.0%, 87.0%, and 80.0% accuracy against the gold standard specifically. Meanwhile, instruction-tuned GPT-4o achieves 94.0% agreement against the same gold standard.

Table 2: Agreement for BioMistral-7B open-ended evaluation. Gold standard = 100 sampled questions agreed by human reviewers.

Judge	Agreement
Reviewer A	89.0%
Reviewer B	87.0%
Reviewer C	80.0%
GPT-4o	94.0%

The GPT-4o confusion-matrix analysis used for calibration is provided in supplementary materials.

Table 3: Calibrated accuracy (%) and p-values for zero-shot vs. best ICL performance across five models using the Med-HEAL pipeline.

Model	Zero-shot	Med-HEAL	Δ	p
BioMistral-7B	48.1 ± 5.0	50.4 ± 3.3	+2.3	0.11
DeepSeek-R1-8B	76.9 ± 1.3	85.9 ± 0.9	+9.0	< 0.001
Llama-3.1-8B	92.1 ± 1.1	94.1 ± 1.6	+2.0	0.024
Qwen2.5-7B	91.6 ± 1.7	92.8 ± 0.5	+1.2	0.02
Qwen3-8B	96.3 ± 1.3	98.0 ± 1.7	+1.7	0.07

3.3 In-Context Learning for Mitigating Hallucination

We evaluate whether the Med-HEAL dataset can mitigate hallucinations through retrieval-augmented in-context learning (ICL). Experiments are conducted on five open-source models used in clinical environments: BioMistral-7B, DeepSeek-R1-Distill-8B, Qwen2.5-7B, Llama-3.1-8B, and Qwen3-8B. Since each summary generated by an LLM still needs a judge to determine whether it matches the ground-truth summary, we continue to use GPT-4o as the judge and report calibrated results. To account for judge noise, we apply Rogan-Gladen calibration [20] using GPT-4o's estimated true positive rate (TPR) and false positive rate (FPR) on the human-verified subset:

$$p = \frac{\hat{p} - \text{FPR}}{\text{TPR} - \text{FPR}},$$

where $\hat{p}$ is the observed GPT-4o-judged accuracy and p is the calibrated accuracy.

Table 3 shows calibrated Med-HEAL results against zero-shot. All five models improve (1%–9%), with significant gains ($p < 0.05$) for DeepSeek, Llama3, and Qwen2.5. DeepSeek has the largest gain (+9.0%), while Qwen3 shows a smaller near-ceiling gain (+1.7%, $p = 0.07$).

BioMistral does not benefit significantly from self-critique, likely due to limited effective context budget and weaker instruction-following for self-diagnosis in long-note settings.

Effect of Med-HEAL Pipeline. Detailed strategy-by-strategy comparison and discussion are provided in supplementary materials.

4 Discussion

This work introduces Med-HEAL, a framework for systematically collecting clinically realistic hallucination cases and mitigating them through a self-critique regeneration pipeline.

Our experiments across five open-source models reveal several critical patterns. First, naive retrieval-based ICL strategies provide little or no improvement over zero-shot performance and sometimes degrade accuracy. These results suggest that simply retrieving semantically similar demonstrations does not reliably guide models to avoid hallucinations in complex clinical reasoning tasks.

Second, Chain-of-Thought prompting is model-dependent and does not consistently reduce hallucinations across models.

In contrast, the Med-HEAL self-critique pipeline provides more consistent gains, especially for mid-performing models, while near-ceiling models such as Qwen3 show smaller improvements.

These results suggest that asking more capable models to explicitly examine and revise their own reasoning, with support from correct and hallucination examples, can effectively mitigate some hallucinations in EHR-based question answering.

Several limitations remain. First, the current Med-HEAL dataset is generated primarily using BioMistral-7B as the source model, which may limit the diversity of hallucination patterns captured. Expanding the dataset to include hallucinations generated by multiple models would improve coverage of possible failure modes. Meanwhile, we can fully investigate and explore the reliability of LLM-as-a-judge. Second, although the dataset includes rich annotations from both LLM judges and human reviewers, our current experiments mainly leverage correctness labels and example responses. Future work may explore alternative ways to utilize these annotations, including supervised fine-tuning, preference learning, or taxonomy-aware prompting strategies. Finally, model accuracy decreases substantially as the number of discharge notes increases, indicating that variation in clinical context length should be investigated in greater detail.

Overall, Med-HEAL demonstrates that hallucination-aware datasets, combined with self-critique prompting, provide a practical approach to improving the reliability of medical LLMs, particularly in tasks involving complex clinical narratives.

5 Conclusion

This paper presents Med-HEAL, a framework for studying and mitigating hallucinations in medical LLMs through clinically grounded evaluation and hallucination-aware prompting. Our experiments demonstrate that conventional retrieval-based in-context learning offers little benefit for this task, whereas a self-critique pipeline consistently improves performance across multiple open-source models. Combining self-critique with targeted demonstrations further improves accuracy by providing examples that address predicted reasoning failures.

These findings suggest that enabling models to identify and correct their own reasoning errors is a promising direction for improving the reliability of medical LLMs. Med-HEAL provides both a dataset and a practical self-critique in-context learning strategy for advancing safer clinical AI systems.

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